The Gram double pencil: a spectral census for the zeros of ζ, and an Euler-anchored regulator

2026-08-14

Abstract

We attach to each window of the critical line a pair of Hankel matrices computed by contour integration of ξ'/ξ, and prove that the number of zeros off the critical line in the window equals the negative inertia of the first matrix (Theorem 2); the remaining spectral data of the pair counts the zeros with multiplicity, their distinct support, and their ordinates (Theorem 5). Both statements are unconditional. The Riemann Hypothesis is thereby equivalent to a Stieltjes-type positivity condition, window by window, in a quotient chart obtained from the functional equation (Proposition 6). We then construct, in the half plane of absolute convergence and with no reference to zeros, an Euler-anchored double pencil which is unconditionally positive semidefinite (Proposition 8), and prove that its spectrum can be transported isospectrally by an explicit finite-dimensional regulator whenever a bank response matrix is nonsingular (Theorem 9), extended through spectral collisions by a block regulator (Lemma 11). We then construct the readout itself: the warp — the flow of a bank-computable velocity field — exists, is real and order-preserving, and carries the frozen labels exactly onto the moving spectrum (Theorem 13); the characteristic polynomial obeys an exact determinant-pullback law with nonvanishing cofactor (Theorem 14). The program thereby reduces to a single hypothesis, the seat (Hypothesis 29): terminal seating of the warped anchor spectrum on the window support, from which the positivity, the census and the critical line follow (Theorem 36). The remaining obligations are stated precisely in §9.

1 Setup

Let ξ be the completed Riemann zeta function and put

$$A(z) := \xi\left(\tfrac{1}{2} - iz\right),$$

an entire function of order one, even ($A(-z) = A(z)$, the functional equation), real on $\mathbb{R}$, whose zeros are $z_\rho = \gamma_\rho + i(\frac{1}{2} - \beta_\rho) \cdot (-1)$ for $\rho = \beta_\rho + i\gamma_\rho$ a nontrivial zero of ζ; in particular

$$\rho \text{ on the critical line} \iff z_\rho \in \mathbb{R}.$$

Nonreal zeros of A occur in conjugate pairs $z, \bar{z}$ with equal multiplicities (conjugate symmetry of ξ on the real axis).

Definition 1 (window moments)**.** Let $W = (a, b)$ with A nonvanishing on $\partial W \times \{|y| \leq Y\}$, $Y > \frac{1}{2}$, and let ∂R_W be the positively oriented rectangle boundary. Put

$$\mu_k(W) := \frac{1}{2\pi i} \oint_{\partial R_W} z^k \, \frac{A'(z)}{A(z)} \, dz, \qquad k \geq 0,$$

and for $n \geq 1$,

$$H_n(W) := \big(\mu_{\alpha+\beta}(W)\big)_{\alpha,\beta=0}^{n-1}, \qquad H_n^{(1)}(W) := \big(\mu_{\alpha+\beta+1}(W)\big)_{\alpha,\beta=0}^{n-1}.$$

By the argument principle, with $x_1, \ldots, x_r$ the distinct real zeros of A in the window (multiplicities $a_i \geq 1$) and $z_j, \bar{z}_j$, $j = 1, \ldots, q$, the distinct conjugate pairs (equal multiplicities $b_j \geq 1$),

$$\mu_k(W) \;=\; \sum_{i=1}^{r} a_i x_i^k \;+\; \sum_{j=1}^{q} b_j \big(z_j^k + \bar{z}_j^k\big). \tag{1}$$

All $\mu_k(W)$ are real and computable from A alone; no knowledge of the zeros is used. Write $m := r + 2q$ for the number of distinct support points.

2 The inertia identity

Theorem 2 (inertia)**.** *For every $n \geq m$,*

$$\operatorname{inertia} H_n(W) \;=\; \big(r + q,\; q,\; n - r - 2q\big),$$

where inertia *denotes (positive, negative, zero) eigenvalue counts. In particular*

$$n_-\big(H_n(W)\big) \;=\; q, \qquad \textit{and} \qquad H_n(W) \succeq 0 \iff q = 0,$$

i.e. the negative inertia counts exactly the off-critical-line conjugate pairs.

Proof. Let $v(t) := (1, t, \ldots, t^{n-1})^T$. By (1),

$$H_n \;=\; \sum_{i=1}^{r} a_i\, v(x_i) v(x_i)^T \;+\; \sum_{j=1}^{q} b_j \Big[v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T\Big].$$

Write $v(z_j) = A_j + iB_j$ with $A_j, B_j \in \mathbb{R}^n$; then $v(\bar{z}_j) = A_j - iB_j$ and

$$v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T \;=\; 2A_j A_j^T - 2B_j B_j^T, \tag{2}$$

the minus sign being the whole mechanism. Hence $H_n = UJU^T$ with

$$U = \big[v(x_1) \cdots v(x_r)\; A_1\, B_1 \cdots A_q\, B_q\big], \qquad J = \operatorname{diag}(a_1, \ldots, a_r,\, 2b_1, -2b_1, \ldots, 2b_q, -2b_q),$$

so inertia $J = (r + q, q, 0)$.

U has full column rank m: the complex matrix $V = [v(x_1), \ldots, v(x_r), v(z_1), v(\bar{z}_1), \ldots]$ is Vandermonde on m distinct nodes with $n \geq m$, so its leading $m \times m$ minor equals $\prod_{\alpha<\beta}(s_\beta - s_\alpha) \neq 0$ and $\operatorname{rank} V = m$; replacing each pair $(v(z_j), v(\bar{z}_j))$ by $(A_j, B_j) = \big(\frac{v(z_j)+v(\bar{z}_j)}{2}, \frac{v(z_j)-v(\bar{z}_j)}{2i}\big)$ is an invertible change of columns, so $\operatorname{rank} U = m$.

Take a thin QR factorisation $U = QR$ with $Q^T Q = I_m$ and R invertible, and extend Q to an orthogonal $\widetilde{Q} = [Q, Q_\perp]$. Then

$$\widetilde{Q}^T H_n \widetilde{Q} = \begin{pmatrix} RJR^T & 0 \\ 0 & 0_{n-m} \end{pmatrix},$$

and by Sylvester's law of inertia inertia$(RJR^T) =$ inertia(J). The zero block contributes $n - m$ null directions. $\square$

Remark 3. The hypothesis $n \geq m$ is necessary: an off-line pair is invisible to a Hankel matrix smaller than the distinct support count. Since $\mu_0(W)$ is the number of zeros in W with multiplicity and $m \leq \mu_0(W)$, the choice $n := \lceil \mu_0(W) \rceil + 1$ is admissible and computable from the bank alone.

Remark 4 (collisions). If a conjugate pair approaches $\mathbb{R}$, the corresponding negative eigenvalue tends to 0^- and at collision becomes a null direction; the invariant through such an event is the inertia, not any individually ordered eigenvalue. Consequently the correct target is non-negativity $H_n \succeq 0$ (equivalently $n_- = 0$), never strict positivity: the $n - m$ structural nulls and the collision nulls are expected.

3 The census, and the equivalent form of RH

Theorem 5 (spectral census)**.** *Let $n \geq m$. From the spectral data of $H_n(W)$ and the value $\mu_0(W)$:*

$$\mu_0 = \textit{number of zeros in } W \textit{ with multiplicity}, \qquad \operatorname{rank} H_n = m = r + 2q,$$

$$n_-(H_n) = q, \qquad n_+(H_n) - n_-(H_n) = r,$$

and $\mu_0 > \sum_i 1 + \sum_j 2$ (i.e. μ_0 exceeds the distinct count) if and only if some zero in W is multiple. Moreover the generalized eigenvalues of the pencil $\left(H_n^{(1)}, H_n\right)$ restricted to $\operatorname{ran} H_n$ *are exactly the support points x_i and $z_j, \bar{z}_j$.*

Proof. The first four items are Theorem 2 together with $\mu_0 = \sum a_i + 2 \sum b_j$. For the pencil statement, $H_n = UJU^T$ and $H_n^{(1)} = UJ\Sigma U^T$ where Σ is the diagonal matrix of support points in the basis of columns of U (Vandermonde shift identity $v(t)^T \cdot t =$ shifted moments); on $\operatorname{ran} H_n$ the pencil is therefore similar to Σ. This is Gaussian quadrature for the measure (1). □

Proposition 6 (quotient chart; Stieltjes form)**.** *Let $w = z^2$ and $\nu_k(W) := \mu_{2k}(W)$. Under $z \mapsto w$: a real zero maps to $w > 0$; a conjugate pair $x \pm iy$ ($x \neq 0$) maps to a conjugate pair in w; a pair on the imaginary axis maps to $w < 0$. Consequently, for n large enough in the sense of Theorem 2,*

$$\textit{all zeros of } A \textit{ in } W \textit{ are real} \iff \left(\nu_{\alpha+\beta}\right) \succeq 0 \;\; \textit{and} \;\; \left(\nu_{\alpha+\beta+1}\right) \succeq 0,$$

the Stieltjes condition for the measure $\sum_\rho m_\rho \delta_{z_\rho^2}$. Hence RH is equivalent to: for every window of a tiling of $\mathbb{R}$, both Hankel matrices of the even moments are positive semidefinite.

Proof. The functional equation makes A even, so the multiset of zeros is symmetric under $z \mapsto -z$ and descends to the w-chart. Reality and non-negativity of the w-support is exactly the Stieltjes moment condition (Hamburger positivity of both $(\nu_{\alpha+\beta})$ and its shift). The equivalence with the location statement is Theorem 2 applied in the w-chart together with the support map. □

Corollary 7 (sign rigidity)**.** *Let $t \mapsto A_t$ be a continuous deformation with zeros varying continuously and no zero crossing ∂W. Then $H_n(t)$ is continuous, and a conjugate pair is created or annihilated at t_0 if and only if an eigenvalue of $H_n(t)$ crosses 0 at t_0; at simultaneous collisions the invariant is the inertia change. Consequently, if $H_n(t) \succeq 0$ for all t, then $n_-(H_n(t)) \equiv 0$ and no off-line pair is ever born. If moreover $H_n(t_*) \succeq 0$ is known at one parameter t_* (for instance in a numerically verified range), the conclusion propagates along the deformation.*

4 Unconditional statements

The following hold with no hypothesis.

(U1) For every window W and $n \geq m(W)$: $q(W) = n_-\big(H_n(W)\big)$; the off-critical-line pair count of a window is the negative inertia of a matrix computed by contour integration of A'/A.

(U2) The census of Theorem 5 holds as stated.

(U3) RH is equivalent to the Stieltjes condition of Proposition 6 on every window of a tiling, equivalently (Corollary 7) to the absence of a negative-sector eigenvalue crossing under the registered deformation.

(U4) For every window contained in the numerically verified range ($|t| \leq 3 \cdot 10^{12}$ by the computations of Platt and Trudgian [3]), $q(W) = 0$ and hence $H_n(W) \succeq 0$: the positivity hypothesis is an established fact on an initial segment of the tiling.

(U5) With $Y > \frac{1}{2}$ the top edge of ∂R_W lies in the half plane of absolute convergence and the bottom edge maps there by the functional equation, giving a decomposition $H_n = H^{\mathrm{pr}} + H^{\mathrm{arch}} + H^{\mathrm{side}}$ into blocks computable from Dirichlet data, Γ-factors and the lateral edges respectively; in a tiling, adjacent lateral contributions cancel pairwise.

5 The Euler-anchored pencil and its regulator

Nothing in this section refers to zeros; the dependency order is

$$\text{Euler/FE bank} \;\to\; (G_1, G_0) \;\to\; \text{real spectrum} \;\to\; \text{contour transport} \;\to\; \text{residues} \;\to\; \text{zeros},$$

and no step may be inverted.

Proposition 8 (Euler anchor)**.** *Fix $s_0 > 1$ and $N \geq 1$, and set*

$$G_\ell[j,k] \;:=\; \sum_{n\geq 2} \Lambda(n)\, n^{-s_0} \,(\log n)^{\,j+k+\ell}, \qquad 0 \leq j,k < N, \quad \ell \in \{0,1\}.$$

The series converge absolutely, and G_0, G_1 are the Hankel matrices of the positive measure $\sum_n \Lambda(n) n^{-s_0} \delta_{\log n}$ supported in $[\log 2, \infty)$. Hence unconditionally

$$G_0 \succeq 0, \qquad G_1 \succeq 0,$$

and $\operatorname{spec}(G_1, G_0) \subset [\log 2, \infty)$ consists of the Gauss quadrature nodes of that measure.

Proof. Absolute convergence for $s_0 > 1$ is standard. Positivity is the Hamburger/Stieltjes criterion for a positive measure on $[\log 2, \infty)$: for $c \in \mathbb{R}^N$ and $P(x) = \sum_j c_j x^j$, $c^T G_0 c = \int P^2 \, d\sigma \geq 0$ and $c^T G_1 c = \int x P^2 \, d\sigma \geq 0$. $\square$

Theorem 9 (unconditional reverb regulator)**.** *Let $t \mapsto (G_0(t), G_1(t))$ be a smooth family with $G_0(t) > 0$ on the active quotient, and set $S(t) := G_0^{-1/2} G_1 G_0^{-1/2}$, so $S = S^*$ and* $\operatorname{spec} S =$

spec(G_1, G_0). *Suppose the spectrum of $S(t_0)$ is simple, with spectral projections P_a and eigenvectors v_a, and let the transport be controlled,*

$$\dot{S} \;=\; F(u) \;=\; F_0 + \sum_{r=1}^{m} u_r F_r, \qquad F_r = F_r^*.$$

Define $M_{ar} := \langle v_a, F_r v_a \rangle$ and $b_a := -\langle v_a, F_0 v_a \rangle$. If $\det M \neq 0$, then $u := M^{-1} b$ is the unique control with $\Delta\big(F(u)\big) := \sum_a P_a F(u) P_a = 0$, and with

$$K := \sum_{a \neq b} \frac{P_a F(u) P_b}{\lambda_a - \lambda_b}, \qquad K^* = -K,$$

one has $\dot{S} = [S, K]$. Consequently $S(t) = U(t) S(t_0) U(t)^*$ with $\dot{U} = -KU$, U unitary, and

$$\operatorname{spec} S(t) \;=\; \operatorname{spec} S(t_0) \quad \textit{exactly.}$$

In particular the transported pencil remains Hermitian-definite and its generalized spectrum remains real.

Proof. First-order perturbation theory gives $\dot{\lambda}_a = \langle v_a, F v_a \rangle$, so $\Delta(F)$ is precisely the component of the flow able to move eigenvalues; the linear system $Mu = b$ expresses $\Delta(F(u)) = 0$ and is uniquely solvable iff $\det M \neq 0$. For such u, using $\sum_a P_a = I$,

$$[S, K] = \sum_{a \neq b} (\lambda_a - \lambda_b) \frac{P_a F(u) P_b}{\lambda_a - \lambda_b} = \sum_{a \neq b} P_a F(u) P_b = F(u) - \Delta(F(u)) = F(u) = \dot{S}.$$

$K^* = -K$ since $F(u)^* = F(u)$ and the denominators are real and antisymmetric in (a, b). The solution of $\dot{U} = -KU$, $U(t_0) = I$, is unitary, and $\frac{d}{dt}\big(U^* S U\big) = U^*\big(\dot{S} - [S, K]\big) U = 0$. □

Remark 10 (numerical verification). For the Euler anchor of Proposition 8 with $s_0 = 1.5$, $N = 4$, control channels given by reweighting the prime channels $p \in \{2, 3, 5, 7\}$ and raw flow $F_0 = \partial_{s_0} S$: the spectrum is simple $(0.9619, 2.7211, 5.6399, 8.8009)$, $\det M = -0.128$ with $\operatorname{rank} M = 4$ and condition number $3.4 \cdot 10^2$; the solved control annihilates all diagonal drifts to 10^{-12}, and $\|[S, K] - (F - \Delta(F))\| = 1.3 \cdot 10^{-11}$ with $\|K + K^T\| = 1.7 \cdot 10^{-13}$.

Lemma 11 (block regulator through collisions)**.** *Let $S = S^*$ have distinct eigenvalues Λ_α with spectral projections P_α of arbitrary finite multiplicities, and let the flow be controlled as in Theorem 9, $\dot{S} = F(u) = F_0 + \sum_r u_r F_r$, $F_r = F_r^*$. With*

$$\Delta(F) := \sum_\alpha P_\alpha F P_\alpha, \qquad K := \sum_{\alpha \neq \beta} \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta},$$

one has $[S, K] = F - \Delta(F)$ and $K^* = -K$. To first order the group Λ_α splits by the spectrum of the compression $P_\alpha F P_\alpha$; hence the regulated condition is the block equation $P_\alpha F(u) P_\alpha = 0$ for all α, a linear system in u generalizing $Mu = b$, and when it is solvable the flow $\dot{S} = [S, K]$ is isospectral with multiplicities. Between collisions Theorem 9 applies; across a collision the invariant is the inertia (Corollary 7); the regulator therefore extends piecewise along any transport with finitely many collision events.

Proof. $SP_\alpha = \Lambda_\alpha P_\alpha$ and $\sum_\alpha P_\alpha = I$ give

$$[S, K] = \sum_{\alpha \neq \beta} (\Lambda_\alpha - \Lambda_\beta) \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta} = F - \Delta(F),$$

and $K^* = -K$ since $F^* = F$ and the real denominators are antisymmetric in (α, β). The first-order motion of a degenerate group is degenerate perturbation theory: the group splits by the eigenvalues of $P_\alpha F P_\alpha$, which vanish identically iff the block equation holds; the condition is linear in u, and on solution intervals the Lax argument of Theorem 9 is unchanged. □

6 Warp covariance: the bank-generated readout

Theorem 9 fixes the abstract spectrum $\{\lambda_a\}$ of the regulated flow; the same transport, unregulated, moves it. The first chart makes reality manifest, the second carries the physical ordinates, and the map between them — the warp — is not a hypothesis to be assumed but an object the bank constructs. Throughout this section $t \mapsto (G_0(t), G_1(t))$ is a smooth transport of the continued jet pair, whose inputs are values and derivatives of the continued logarithmic derivative along the path — function data; no zero location enters — and the parameter interval is *regular*: $G_0(t) > 0$ on the active quotient and $\operatorname{spec} S_t$ simple, where $S_t := G_0^{-1/2} G_1 G_0^{-1/2}$. The finitely many failures of regularity are the collision and rank-drop events, crossed piecewise by Lemma 11.

Definition 12 (bank velocity field; warp)**.** Let $\lambda_1(t) < \cdots < \lambda_n(t)$ be the eigenvalues of S_t, realized as the generalized eigenpairs $G_1 w_a = \lambda_a G_0 w_a$ normalized by $w_a^T G_0 w_a = 1$, and set

$$d_a(t) := w_a^T \big(\dot G_1 - \lambda_a \dot G_0\big) w_a \qquad \big(= \langle v_a, \dot S_t\, v_a\rangle,\ v_a = G_0^{1/2} w_a\big).$$

Let V_t be the unique real polynomial of degree $\leq n-1$ with $V_t(\lambda_a(t)) = d_a(t)$, and let the *warp* φ_t be the flow of the scalar equation $\dot x = V_t(x)$, $\varphi_0 = \mathrm{id}$.

Theorem 13 (warp existence and exact covariance)**.** *On every regular interval the warp is defined on a neighborhood of* $[\lambda_1(0), \lambda_n(0)]$ *and satisfies:*

(i) ***bank determination:*** *V_t, hence φ_t, is computed from $(G_0, G_1)(t)$ and $(\dot G_0, \dot G_1)(t)$ alone;*

(ii) ***reality and order:*** *$\varphi_t(\mathbb{R} \cap \mathrm{dom}) \subseteq \mathbb{R}$ and $x \mapsto \varphi_t(x)$ is strictly increasing, in particular injective;*

(iii) ***exact covariance:*** *$\varphi_t(\lambda_a(0)) = \lambda_a(t)$ for every a and every t in the interval.*

Proof. (i) is the construction. For (ii): V_t has real coefficients — real nodes, real diagonal drifts of a Hermitian family — so the flow preserves $\mathbb{R}$; the field is polynomial in x with coefficients continuous in t, hence locally Lipschitz in x uniformly on compact t-intervals, so distinct scalar characteristics never cross and φ_t is strictly increasing where defined. For (iii): first-order perturbation theory for the Hermitian-definite pencil gives $\dot\lambda_a = w_a^T(\dot G_1 - \lambda_a \dot G_0) w_a = d_a(t) = V_t(\lambda_a(t))$, so the eigenvalue path $t \mapsto \lambda_a(t)$ solves the initial-value problem defining $\varphi_t(\lambda_a(0))$, and uniqueness makes them equal. Existence: the node characteristics are the globally defined eigenvalue paths; a point between consecutive nodes is trapped between two of them by (ii); and continuous dependence on initial conditions extends the flow to a neighborhood of the compact hull for the same times. □

Theorem 14 (determinant pullback)**.** *Let $p_t(x) := \det(S_t - xI)$, and let I be an open real interval containing the time-0 active spectrum on which the warp is defined. Then on $\varphi_t(I)$,*

$$p_t \;=\; J_t \cdot \big(p_0 \circ \varphi_t^{-1}\big),$$

with J_t continuous and nonvanishing; consequently

$$Z(p_t) \cap \varphi_t(I) \;=\; \varphi_t\big(\operatorname{spec} S_0\big) \;\subset\; \mathbb{R}.$$

Proof. The field is polynomial in x, so φ_t is a strictly increasing C^1 diffeomorphism of I onto $\varphi_t(I)$. On $\varphi_t(I)$ the functions p_t and $p_0 \circ \varphi_t^{-1}$ have the same zero set $\{\varphi_t(\lambda_a(0))\}_a = \{\lambda_a(t)\}_a$ (covariance and injectivity, Theorem 13), each zero simple (simple spectrum on a regular interval). The ratio $J_t := p_t/(p_0 \circ \varphi_t^{-1})$ is continuous and nonvanishing off the common zeros and extends across each with the nonzero limit $p_t'(\lambda_a(t))/\big(p_0 \circ \varphi_t^{-1}\big)'(\lambda_a(t))$. □

Remark 15 (the two architectures, reconciled). The regulated and unregulated flows describe one transport in two charts: labels frozen with a moving readout, or a moving spectrum read through the identity; by Theorem 13 the whole discrepancy is φ_t. This corrects the former statement of obligation (N), which asked for a spectral equivalence $\det(G_1 - \lambda G_0) = C \det(H_n^{(1)} - \lambda H_n)$ with one spectral variable λ and a constant C: under the regulated flow that identity is unsatisfiable — the transported spectrum stays at the anchor nodes while the window spectrum sits at the ordinates, disjoint multisets — and it is replaced by the pullback identity of Theorem 14, in which the change of spectral variable is explicit and bank-generated.

Proposition 16 (the warp is internally generated on the anchor segment)**.** *On the segment $s_0 > 1$ the transport is $\dot\sigma = -x\,\sigma$ (each prime clock damps by its own frequency), so $\dot m_k = -m_{k+1}$: the moment flow is the pencil's own shift. Let (λ_a, w_a) be the Gauss data of the n-point pencil, P_n the monic node polynomial, and $h_n := \int P_n^2\, d\sigma = m_{2n} - \sum_a w_a \lambda_a^{2n}$ the quadrature error of x^{2n}. Then*

$$\frac{d\lambda_a}{ds_0} \;=\; -\,\frac{h_n}{w_a\, P_n'(\lambda_a)^2}\,,$$

strictly negative for every a. Every quantity on the right except h_n is internal to the pencil; the entire external input of the flow is the single boundary scalar h_n.

Proof. The n-point Gauss data matches the moments m_k for $k \le 2n-1$. Differentiating the matching conditions along $\dot m_k = -m_{k+1}$, the homogeneous solution $\dot w_a = -\lambda_a w_a$, $\dot\lambda_a = 0$ accounts for $-m_{k+1}$ within quadrature exactness, leaving the forcing $-h_n \delta_{k,2n-1}$ at the single degree where exactness fails. The correction $(u_a, \dot\lambda_a)$ solves $\sum_a [u_a Q(\lambda_a) + w_a \dot\lambda_a Q'(\lambda_a)] = -h_n \cdot [x^{2n-1}]Q$ for all Q of degree $\le 2n-1$. Take $Q = \ell_a P_n$ with ℓ_a the Lagrange basis: Q vanishes at every node, $Q'(\lambda_b) = \delta_{ab} P_n'(\lambda_a)$, and $[x^{2n-1}]Q = 1/P_n'(\lambda_a)$; the display follows. Positivity of σ on the segment gives $h_n > 0$, $w_a > 0$. □

Remark 17 (self-calibration, and its limit). Proposition 16 is the Toda-deformation structure of exponentially damped measures: the infinite operator is isospectral — its spectrum is the fixed prime-clock support $\{\log n\}$, the carrier — while the finite pencil's nodes, the readout, move only through the boundary forcing h_n. Verified (`tmp/att205_selfcal_probe.py`, $s_0 = 1.5$, $N = 4$): the internal law matches the direct perturbation drifts to $1.6\cdot 10^{-11}$, with $h_n = 2.2\cdot 10^5$ cancelling against Christoffel factors of order 10^{-4} exactly. Two consequences and one boundary. First, the warp needs no external support on the anchor segment beyond one scalar per instant; the continuation's defects (pole, Γ, window) must enter the finite system through this single boundary channel, so Hypothesis 29 becomes a statement about the behavior of a one-dimensional boundary sequence rather than an $n \times n$ identity. Second, $h_n = \int P_n^2 \, d\sigma$ is manifestly nonnegative exactly as long as the driving functional is a positive measure: where the continuation costs h_n its manifest sign is where the seat's content begins. The boundary: a self-calibrating flow is not a self-certifying one — the device generates its own motion, but the terminal seating certificate must still come from the crossing analysis of Lemmas 22 and 25. Finally, an identity rather than an analogy: for Hankel pairs, $h_n = \det H_{n+1} / \det H_n$, so the boundary scalar that drives the flow *is* the inertia-increment detector of the census — its loss of sign at one size and the appearance of a negative direction at the next are the same fact (verified along the continuation: sign flip at $n = 14$, inertia event at $n = 15$; `tmp/att211_boundary_onset.py`).

Remark 18 (numerical verification). On the convergent segment ($s_0\colon 1.5 \to 1.2$, $N = 4$, exact jets of $-\zeta'/\zeta$; `tmp/att201_warp_probe.py`): the perturbation drift matches finite differences of the eigenvalue paths to $8.9 \cdot 10^{-8}$; the flow of the interpolated field carries the anchor nodes (1.1916, 4.0395, 9.6214, 19.3384) onto the terminal eigenvalues (2.1714, 9.2832, 23.2371, 47.5293) with maximal error $1.9 \cdot 10^{-10}$; an interior test point remains strictly interior. The exact-jet anchor nodes differ from those quoted after Theorem 9, which realize the truncated prime sum at finite cutoff; the two banks are distinct legitimate realizations, and the drift between them is the truncation observation of the ledger.

7 The conditional theorem

Hypothesis 19 (PSD)**.** For every window W of a tiling of $\mathbb{R}$ and $n = \lceil \mu_0(W) \rceil + 1$, $H_n(W) \succeq 0$; equivalently (Proposition 6) both Hankel matrices of the even moments are positive semidefinite in the quotient chart.

Theorem 20. *Assume Hypothesis 19. Then*

1. *every nontrivial zero of ζ lies on the critical line;*
2. *for each window the spectral data of $H_n(W)$ returns the full census of Theorem 5, with $n_- \equiv 0$;*
3. *the zeros are the real generalized eigenvalues of the explicitly computed Hermitian pencils $\big(H_n^{(1)}(W), H_n(W)\big)$.*

Proof. By Theorem 2, $H_n(W) \succeq 0$ with $n \geq m(W)$ forces $q(W) = 0$: no conjugate pair of zeros of A lies in W. Nonreal zeros occur in conjugate pairs with equal real part, hence both members lie in the same window of the tiling (for a pair on a boundary, apply the argument to a shifted tiling). Every nontrivial zero of ζ corresponds to a zero of A lying in some window, and only finitely many per window by Riemann–von Mangoldt. Hence all zeros of A are real, i.e. all nontrivial zeros of ζ have real part $\frac{1}{2}$. Items (2) and (3) are Theorem 5, whose pencil is Hermitian-definite on $\operatorname{ran} H_n$ under Hypothesis 19. $\square$

8 The seat: reduction to a single hypothesis

Definition 21 (terminal defect)**.** Fix a window W with matched dimension n and a transport path whose terminal chart is the window chart, and let $(G_0(1), G_1(1))$ be the terminal transported pair. Put

$$D_0 := G_0(1) - H_n(W), \qquad D_1 := G_1(1) - H_n^{(1)}(W).$$

Both differences are computed from function data — jets of the continued logarithmic derivative on one side, contour integrals of A'/A on the other — so the defect is defined with no reference to zeros. Obligation (T) of §9 evaluates D_ℓ in closed form; the smooth blocks are now explicit. The pole block is the Hankel of the exponential measure $e^{-(s_0-1)x}dx$ on $[0,\infty)$, moments $k!/(s_0-1)^{k+1}$, positive semidefinite on the convergent segment $s_0 > 1$. The Γ-block (trivial zeros) is the Hankel of the measure $e^{-(s_0+2)x}\big(1-e^{-2x}\big)^{-1}dx$ on $(0,\infty)$, with moments in closed Hurwitz form

$$k!\; 2^{-(k+1)}\,\zeta\big(k+1,\; 1+\tfrac{s_0}{2}\big) \qquad (k \geq 1),$$

a manifestly positive density for every $s_0 > -2$, hence positive semidefinite *throughout the strip.* The entire smooth part of the defect therefore adds positivity at every station of the transport; only the window and lateral blocks of (T) remain unevaluated. (These closed forms are the trivial-zero and pole components of the jet engine used in the numerical maps, validated there against direct differentiation to $5\cdot 10^{-5}$.)

Lemma 22 (null-cone flow decomposition). *Let $\sigma = \sum_{x\in X} w(x)\delta_x$ be an atomic functional on a conjugation-closed multiset $X = \overline{X}$ with weights satisfying $w(\bar{x}) = \overline{w(x)}$, and let H be the Hankel matrix of its moments, of size n. For every real polynomial P of degree $< n$ with coefficient vector v,*

$$v^T H v \;=\; \sum_{x\in X\cap\mathbb{R}} w(x)P(x)^2 \;+\; \sum_{\{z,\bar{z}\}\subset X\setminus\mathbb{R}} 2\,\mathrm{Re}\big[w(z)P(z)^2\big],$$

in which the real-support sum is termwise nonnegative whenever $w \geq 0$ on $X\cap\mathbb{R}$, while the conjugate-pair terms are unrestricted in sign. Consequently, along a weight homotopy $t \mapsto w_t$ with $\dot{w}_t \geq 0$ on $X\cap\mathbb{R}$, the flow through a null direction ($v^T H_t v = 0$) obeys

$$v^T \dot{H}_t v \;\geq\; \sum_{\{z,\bar{z}\}} 2\,\mathrm{Re}\big[\dot{w}_t(z)P(z)^2\big] :$$

a negative-sector crossing must be fed by conjugate-pair terms; real support cannot produce one. In particular, on any pointwise-monotone kernel homotopy whose weighted support is real and which starts at a nonnegative-weight configuration, $n_-(H_t) \equiv 0$ unconditionally.

Proof. Group the atoms: conjugate pairs combine to $2\mathrm{Re}[w(z)P(z)^2]$ because P is real, and the real-support terms are $w(x)P(x)^2 \geq 0$ termwise. The flow statement is the same identity applied to $\dot{H}_t$, discarding the nonnegative real sum. The last claim is closedness of the PSD cone: positive start, and no crossing is available. □

Remark 23 (measured margin law). On the verified window $W = (10, 32)$ ($n = 5$, four zeros; `tmp/att202_margin_probe.py`), along the error-function smoothing of the window indicator: the dead-direction mass $v_*^T H(\tau)v_*$ (v_* the terminal null polynomial vanishing at the four interior zeros) is monotone with positive flow sign ($+1.4\cdot 10^{-2}$ at $\tau = 1$); by $\tau = 1$ its source is 100% the boundary-nearest exterior zero $\gamma_5 = 32.935$; and $\lambda_{\min}(H(\tau))$ tracks the Rayleigh value $v_*^T H v_*$ to 99.8% at $\tau = 0.25$. The positivity slack of the window pencil near the sharp limit is the weight leaking onto the nearest exterior zero through the terminal null polynomial: the margin has a single computable source.

Remark 24 (cosine transport of the pair-fed flow). The conjugate-pair terms of Lemma 22 have an exact operator form: for F entire and real on $\mathbb{R}$, $F(x+iy)+F(x-iy) = 2\cos(y\partial_x)F(x)$ (Taylor expansion, absolutely convergent). Hence a pair at depth y feeds the flow with the *same* real density $g = \dot{w}\,P^2$ that a real zero would, transported by $\cos(y\partial_x)$. The Fourier multiplier $\cos(y\xi)$ is nonnegative for $|\xi| \le \pi/(2y)$, and $y < \frac{1}{2}$ unconditionally (the critical strip), so the worst-case positivity radius of the transport is the band $|\xi| \le \pi$. Two consequences. First, a quantitative target: pair-fed dominance on a null direction requires the density g to carry Fourier mass beyond the positivity radius of the actual pair depths present — shallow pairs feed like real zeros. Second — correcting an identification staked in an earlier draft of this remark — the cross-check against the narrow-support literature *fails on the numbers, and informatively so*: the classical unconditional positivity radius (Yoshida [4]) is $\log 2 = 0.693$, set by the *prime vacuum* (no Λ-support below the first prime), whereas the cosine-transport radius is $\pi = 3.14$, set by pair depth. The two are distinct mechanisms on opposite sides of the explicit formula: prime-vacuum limits the classical regime, pair depth limits this transport, and $\pi > \log 2$ means the pair-depth constraint is strictly the weaker one — positivity of the pair-fed flow persists beyond the prime-vacuum boundary, and the two constraints compose rather than coincide. The present formulation additionally needs the sign only on null directions (a thinner set) and may weigh actual pair depths against unconditional density estimates rather than assuming the worst case everywhere.

Lemma 25 (resonance necessity)**.** *Let P be a real polynomial of degree d with roots $r_1, \dots, r_d$ (nonreal ones in conjugate pairs), let $z_0 = x_0 + iy_0$ with $y_0 > 0$, and let $w(z_0) \in \mathbb{C}^\times$ with $|\arg w(z_0)| \le \varphi_w < \frac{\pi}{2}$. Define the deviation of each root as the angle between $z_0 - r_j$ and the real line,*

$$\delta_j \;:=\; \arctan\frac{|y_0 - \operatorname{Im} r_j|}{|x_0 - \operatorname{Re} r_j|} \;\in\; (0, \tfrac{\pi}{2}] \qquad (\delta_j := \tfrac{\pi}{2} \text{ if } x_0 = \operatorname{Re} r_j).$$

If $\operatorname{Re}\big[w(z_0)P(z_0)^2\big] \le 0$*, then*

$$\sum_{j=1}^{d} \delta_j \;\ge\; \frac{\pi}{4} - \frac{\varphi_w}{2},$$

and in particular some root satisfies

$$|x_0 - \operatorname{Re} r_j| \;\le\; \frac{|y_0 - \operatorname{Im} r_j|}{\tan\big((\frac{\pi}{4} - \frac{\varphi_w}{2})/d\big)} \;\approx\; \frac{4d}{\pi}\big(y_0 + |\operatorname{Im} r_j|\big) \quad \textit{for small angles.}$$

Proof. Each squared factor of $P(z_0)^2 = c^2\prod_j (z_0 - r_j)^2$ contributes, modulo 2π, an argument $\pm 2\delta_j$: for a factor with argument $\theta_j \in (0,\pi)$, either $\theta_j = \delta_j$ or $\theta_j = \pi - \delta_j$, and $2(\pi - \delta_j) \equiv -2\delta_j$. Hence $\arg\big[P(z_0)^2\big] \equiv \sum_j \pm 2\delta_j \pmod{2\pi}$, and $\big|\sum_j \pm 2\delta_j\big| \le 2\sum_j \delta_j$. A nonpositive real part of $w(z_0)P(z_0)^2$ requires its total argument to lie within $\frac{\pi}{2}$ of π modulo 2π, so $2\sum \delta_j + \varphi_w \ge \frac{\pi}{2}$. The root bound is the pigeonhole $\max_j \delta_j \ge (\frac{\pi}{4} - \frac{\varphi_w}{2})/d$. □

Corollary 26 (separation forces depth)**.** *In the setting of Lemma 25, suppose every root satisfies $|x_0 - \operatorname{Re} r_j| \ge d$ and $|\operatorname{Im} r_j| \le \iota$. Then a nonpositive pair term at (x_0, y_0) requires*

$$y_0 \;\ge\; d\,\tan\frac{\frac{\pi}{4} - \frac{\varphi_w}{2}}{d^\circ P} \;-\; \iota \;\approx\; \frac{\pi d}{4\,d^\circ P} \quad (\varphi_w, \iota \textit{ small}),$$

where $d^\circ P$ is the degree. Equivalently: the abscissas from which a pair of depth y_0 can cross are confined to the $\big((y_0+\iota)/\tan((\frac{\pi}{4} - \frac{\varphi_w}{2})/d^\circ P)\big)$-neighborhood of the null-root abscissa set — a set *of total measure at most* $\approx \frac{8}{\pi}\,(d^\circ P)^2\, y_0$ *— while pairs elsewhere in the window are spectators regardless of the flow.*

Proof. Each deviation obeys $\delta_j \le \arctan((y_0+\iota)/d)$, so $\sum_j \delta_j \le d^\circ P \cdot \arctan((y_0+\iota)/d)$; Lemma 25 requires this to reach $\frac{\pi}{4} - \frac{\varphi_w}{2}$. The neighborhood statement is the same inequality read per root, and the measure bound is $d^\circ P$ intervals of the stated half-width. □

Remark 27 (the composition slot). Corollary 26 is the deterministic half of a two-part exclusion: crossing-capable pairs at depth y are confined to an explicit set of measure $O(n^2 y)$ per window, while the number of pairs of depth $\ge y$ up to height T is counted unconditionally by the zero-density estimates: $N(\sigma, T) \ll T^{3(1-\sigma)/(2-\sigma)+o(1)}$ (Ingham [1]) and $N(\sigma, T) \le T^{30(1-\sigma)/13+o(1)}$ (Guth–Maynard [2], the first exponent improvement in eighty-four years), so pairs at depth y number $\ll T^{30(\frac{1}{2}-y)/13+o(1)}$. Two calibrations. The composition of the two counts against the window census yields rarity statements — windows admitting a resonance are sparse — and rarity is not exclusion: a count cannot reach zero, so this route's unaided yield is a new-mechanism density theorem, not the seat. Closing the seat on a given window requires the per-window exclusion — path steering within the vulnerable set of Corollary 26, or positivity — and that is where the remaining content sits.

Remark 28 (crossings are local resonances). Combining Lemmas 22 and 25: a negative-sector crossing requires a conjugate pair at depth $y_0 < \frac{1}{2}$ whose abscissa lies within $\approx \frac{4(n-1)}{\pi}\, y_0$ of a root cluster of the null polynomial of the accumulated Hankel — with the kernel's phase budget φ_w a design parameter one makes small on the strip. Shallow pairs can cross only in near-collision with a null root (the continuity of Corollary 7, now quantitative); deep pairs have room, but their number is the subject of unconditional zero-density estimates. The seat's remaining content is the exclusion of these localized resonances along one designed path per window; the composition of depth, density and the locality of null directions (the measured margin law) is the open program.

Hypothesis 29 (the seat (S))**.** For every window W of the tiling there is a transport path with finitely many regularity events such that every point of $\operatorname{spec}\big(H_n^{(1)}(W), H_n(W)\big)\big|_{\operatorname{ran} H_n}$ — the window support — is a terminal limit of the warped anchor spectrum $\varphi_t(\operatorname{spec} S_0)$, the remaining $n-m$ directions retiring into the null space through the rank-drop events.

Remark 30 (three contractions of one kernel). The transported objects of this paper are contractions of the single two-variable kernel $B_{jk}(\rho, \rho') = (s_0-\rho)^{-(j+1)}(s_0-\rho')^{-(k+1)}$ along three pairings of the conjugation- and FE-closed zero multiset: the *holomorphic diagonal* (ρ with ρ; the jet-Hankel face, real symmetric), the *functional-equation pairing* (ρ with $1-\rho$; hermitian by conjugation-closure), and the *conjugate pairing* (ρ with $\bar\rho$; a Gram matrix, positive semidefinite always). Atomwise, the FE and conjugate pairings coincide iff $\bar\rho = 1-\rho$, i.e. iff the zero lies on the critical line; the holomorphic diagonal is indefinite for complex frequencies whether or not they lie on the line. Two consequences. First, a calibration: the indefiniteness onset of the jet-Hankel face along the continuation carries no information about zero locations — it is the generic behavior of complex-frequency moment data. Second, the diagnostic content of the program is exactly the coincidence of the FE and conjugate contractions — every zero self-paired by the functional equation is a zero on the line — which is the pairing-language form of the single hypothesis, and the positivity of the FE contraction is its window-free shadow.

Proposition 31 (scalar seat criterion)**.** *Over the nontrivial-zero multiset put*

$$S(s) \;:=\; \sum_{\rho} (s-\rho)^{-1}\,(\bar{s}-1+\rho)^{-1},$$

absolutely convergent for s off the zero set (terms are $O(|\gamma_\rho|^{-2})$), and real-valued: the relabeling $\rho \mapsto 1-\bar{\rho}$, under which the multiset is closed, conjugates its terms. Then

every nontrivial zero lies on the critical line $\iff$ $S(s) \geq 0$ *for all s in the strip off the zero set.*

Proof. ($\Rightarrow$) For ρ on the line, $1-\rho = \bar{\rho}$, so the ρ-term is $(s-\rho)^{-1}\overline{(s-\rho)^{-1}} = |s-\rho|^{-2} > 0$: the sum is termwise positive. ($\Leftarrow$) Let $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac{1}{2}$ and set $s = \rho_0 + \varepsilon e^{i\theta}$. The ρ_0-term is $\varepsilon^{-1}e^{-i\theta}\big((2\beta_0-1)+\varepsilon e^{-i\theta}\big)^{-1}$, of modulus $\sim (\varepsilon|2\beta_0-1|)^{-1}$ with freely rotating phase; choosing θ with $\cos\theta\cdot(2\beta_0-1) < 0$ makes its real part $\to -\infty$ as $\varepsilon \to 0$. Every other term stays bounded on the ε-ball: the remaining members of the ρ_0-family ($\bar{\rho}_0$, $1-\rho_0$, $1-\bar{\rho}_0$) and all other zeros are at distance bounded below, and the full sum of their moduli converges. Hence $S(s) < 0$ for small ε. □

Remark 32 (the traveling seat). Proposition 31 is the $n=1$ member of the FE-paired family $A_{jk}(s) = \sum_\rho (s-\rho)^{-(j+1)}(\bar{s}-1+\rho)^{-(k+1)}$, Hermitian for every anchor s by the same relabeling, with on-line zeros contributing rank-one positive terms at every anchor. Numerically (`tmp/att217_anchor_lift.py`, verified range, $n=8$): as the anchor rises through the strip the form remains positive semidefinite at every station and *seats* on each zero in turn — mass share of the nearest zero 99% at $\tau = \gamma_j$, resolvent spike at each seating, conditioning improving with localization. A negative direction would appear exactly at the seating of an off-line pair: the falsifier is localized in the anchor height.

The source reading required by the working rules identifies the scalar criterion: by partial fractions, $(s-\rho)^{-1}(\bar{s}-1+\rho)^{-1} = (s+\bar{s}-1)^{-1}\big[(s-\rho)^{-1}+(\rho-(1-\bar{s}))^{-1}\big]$, and summing over the paired multiset via the Hadamard expansion and the functional equation ($\xi'/\xi(1-\bar{s}) = -\overline{\xi'/\xi(s)}$) gives the closed form

$$S(s) \;=\; \frac{2\,\mathrm{Re}\,\big[\xi'/\xi(s)\big]}{2\sigma-1}, \qquad \sigma = \mathrm{Re}\,s.$$

Proposition 31 is therefore equivalent to the positivity criterion of Hinkkanen (1997) and Lagarias [5]: RH iff $\mathrm{Re}\,[\xi'/\xi](s) > 0$ for $\sigma > \frac{1}{2}$ — a rediscovery, in FE-paired coordinates, with the pairing trichotomy of Remark 30 supplying a two-line derivation. What is retained beyond the identification: the anchor-lift geometry above (the criterion's falsifier is seat-localized), and the wiring of the criterion into the transport apparatus of this paper. The graded family A_{jk} for $n > 1$ is likewise identified by the required source reading: it is of the Weil–Yoshida hermitian-form species [4, 7] in a resolvent basis at complex anchor — the species carrying Li's criterion as its Laguerre-basis instance (Bombieri–Lagarias [6]) and the Connes–Consani semi-local positivity results as its deepest partial positivity. The bridge runs both ways: partial positivity results for these forms are partial results on the seat, and the transported-pencil apparatus — the internal flow law, the boundary-scalar detector, the seat-localized falsifier geometry — is what this paper adds around the classical species.

Proposition 33 (unconditional positivity on the verified band)**.** *Let T_0 be a height below which every nontrivial zero lies on the critical line ($T_0 = 3\cdot 10^{12}$ by [3]). There is an absolute constant C such that*

$$S(s) \;>\; 0 \qquad \textit{for every } s = \sigma + i\tau \textit{ in the strip with } 30 \leq |\tau| \leq T_0 - C\log T_0.$$

The scalar seat criterion of Proposition 31 therefore holds unconditionally on the full anchor band of the verified range; its open content is confined to anchors above the verification height.

Proof. Split the sum at height T_0. Below: every zero is on the line, so each term is $|s-\rho|^{-2} > 0$; by Riemann–von Mangoldt, $[\,|\tau|, |\tau|+1\,]$ contains a zero once $|\tau| \geq 30$, giving a single-term lower bound $S_<(s) \geq \left(\frac{1}{4}+1\right)^{-1} \cdot \frac{1}{2}$, say. Above: for any zero ρ (on the line or not) at height $\gamma > T_0$, both factors have modulus at least $\gamma - |\tau|$ — $|s-\rho| \geq \gamma - |\tau|$ and $|\bar{s}-1+\rho| \geq \gamma - |\tau|$, since each has imaginary part $\pm(\gamma - |\tau|)$ up to sign conventions — so the term is bounded by $2(\gamma - |\tau|)^{-2}$ in modulus. Integrating against the counting measure,

$$|S_>(s)| \;\leq\; 2\int_{T_0}^{\infty} \frac{dN(\gamma)}{(\gamma - |\tau|)^2} \;\ll\; \frac{\log T_0}{T_0 - |\tau|},$$

which is smaller than the lower bound once $T_0 - |\tau| \geq C \log T_0$ for an absolute C. □

Proposition 34 (band-local seat criterion)**.** *For $H \geq 60$, with C the absolute constant of Proposition 33:*

- *(i) if every zero with $|\gamma| \leq H$ lies on the line, then $S(s) > 0$ for all anchors with $30 \leq |\tau| \leq H - C\log H$ (the proof of Proposition 33 verbatim, with T_0 replaced by H);*
- *(ii) if $S \geq 0$ on the band $|\tau| \leq H$, then every zero with $|\gamma| \leq H-1$ lies on the line (the converse construction of Proposition 31 produces a negative value at an anchor within distance 1 of any off-line zero).*

In particular RH is equivalent to band positivity at every height, with the correspondence exact up to logarithmic boundary layers.

Remark 35 (the induction frame)*.* Proposition 34 places the whole question in one-dimensional coordinates: RH holds iff the band-positivity property is unbounded, and by (U4) it holds below $3 \cdot 10^{12}$. The single remaining statement is the self-extension step — band positivity below H implies it below $H + C\log H$ — and by (ii) this step is equivalent to the absence of a first off-line zero in the boundary layer: necessarily of full strength, as every reduction in this paper must be, but now localized to a moving layer of logarithmic width, where the protecting mass is the verified band below and the threat is a single hypothetical seating just above. The seat, in its minimal coordinates: *the boundary layer never turns.*

Theorem 36 (the chain)**.** *Hypothesis 29 implies Hypothesis 19. Hence, granted the seat alone: every nontrivial zero of ζ lies on the critical line, the census of Theorem 5 is returned on every window, and the zeros are the real spectra of the computed Hermitian pencils (Theorem 20).*

Proof. On every regular interval the transported spectrum is real: the anchor spectrum is real (Proposition 8, S_0 Hermitian) and Theorem 13 transports it by a real map; across regularity events reality is preserved because the pencil remains Hermitian with $G_0 > 0$ on the active quotient (Lemma 11). A limit of reals is real, so by Hypothesis 29 every support point of the window measure (1) is real: $q(W) = 0$ for every window, and Theorem 2 gives $H_n(W) \succeq 0$: Hypothesis 19. □

Remark 37 (necessity, and what has been gained)*.* By the rigidity of the moment problem (§9) any hypothesis implying Hypothesis 19 is equivalent to it; the seat is no exception, and this is forced — it is not a defect of the method. What has changed is the form: of the former obligations, (N), (W1), (W2) and (C) are now theorems (Theorem 9, Lemma 11, Theorems 13 and 14), and the residue is one statement, bank-generated and zero-free in formulation, finite-dimensional on each window.

9 Remaining obligations

Theorem 20 is conditional on Hypothesis 19, and Theorem 36 reduces Hypothesis 19 to the seat. By the rigidity of the moment problem, any pairing whose Gram matrix reproduces the moments $\mu_{j+k}(W)$ *is* the counting pairing of (1), so its positivity is equivalent to the conclusion; a proof must therefore proceed from the bank side, and the precise remaining steps are these.

(T) **Contour transport (defect evaluation).** Evaluate the terminal defect of Definition 21 in closed form,

$$G_\ell[j,k] \;=\; \frac{1}{2\pi i}\oint_\Gamma z^{\,j+k+\ell}\frac{A'(z)}{A(z)}\,dz \;+\; D_\ell[j,k],$$

with D_ℓ the explicit pole, Γ-factor, lateral and windowing contributions of (U5). The chain of Theorem 36 does not consume (T): the defect is already defined as a difference of computed objects; (T) supplies its closed form, the DC structure of the seat, and computational access to Hypothesis 29 on explicit windows.

(S) **The seat.** Hypothesis 29: terminal seating of the warped anchor spectrum on the window support. This is the single remaining hypothesis. By Lemma 22 its content along monotone kernel homotopies is exactly: the conjugate-pair-fed component of the flow never dominates the real-fed component on a null direction of the accumulated Hankel — the only crossing mechanism that exists. The former obligations (N), (W) and (C) are discharged: pencil neutrality by Theorem 9 with the corrected terminal statement of Remark 15; warp covariance — bank determination, reality, injectivity, and the determinant-pullback law — by Theorems 13 and 14; collisions by Lemma 11.

Granted (S), the chain runs forward: the Euler-anchored pencil is positive semidefinite unconditionally (Proposition 8); its spectrum is real and is carried by the bank-generated warp along the transport, exactly and order-preservingly (Theorems 9, 13, 14, Lemma 11); the seat identifies the terminal limits with the window support; reality forces $q \equiv 0$ (Theorem 2); and Theorem 20 concludes.

References

[1] A. E. Ingham, *On the estimation of* $N(\sigma, T)$, Quart. J. Math. Oxford 11 (1940), 291–292.

[2] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, arXiv:2405.20552 (2024).

[3] D. Platt and T. Trudgian, *The Riemann hypothesis is true up to* $3\cdot 10^{12}$, Bull. London Math. Soc. 53 (2021), 792–797.

[4] H. Yoshida, *On hermitian forms attached to zeta functions*, in: Zeta Functions in Geometry, Adv. Stud. Pure Math. 21 (1992).

[5] J. C. Lagarias, *On a positivity property of the Riemann ξ-function*, Acta Arith. 89 (1999), 217–234.

[6] E. Bombieri and J. C. Lagarias, *Complements to Li's criterion for the Riemann hypothesis*, J. Number Theory 77 (1999), 274–287.

[7] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Rend. Mat. Acc. Lincei (9) 11 (2000), 183–233.